Preliminary Project Report

on

AI Driven Predictive Cloud Resource Optimization System

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF
Bachelor of Engineering (Computer Engineering)

BY

Abhay Joshi — (7402)
Anish Kumar Pandey — (7411)
Anish Kumar Pandey — (7412)
Bikash Sharma — (7418)

Under the Guidance of
Prof. (Dr.) Sunil Dhore

Department of Computer Engineering
Army Institute of Technology, Pune 411015.
SAVITRIBAI PHULE PUNE UNIVERSITY
2026-27

Certificate

This is to certify that the Project Entitled **AI Driven Predictive Cloud Resource Optimization System** Submitted by Abhay Joshi (7402), Anish Kumar Pandey (7411), Anish Kumar Pandey (7412), and Bikash Sharma (7418) is a bonafide work carried out by students under the guidance of Prof. (Dr.) Sunil Dhore and is submitted as part of the Bachelor of Engineering (Computer Engineering) Project requirement.

Prof. (Dr.) Sunil Dhore
Internal Guide

Prof. (Dr.) Sunil Dhore
H.O.D

Dr. B P Patil
Principal
Place: AIT, Pune
Date:

Contents

1 Introduction **5**
- 1.1 Details of project work . . . 5
- 1.2 Objectives . . . 6
- 1.3 Scope Of The Project . . . 7
- 1.4 Motivation of the project . . . 7

2 Literature Survey **9**
- 2.1 Production Cloud Resource Management . . . 9
- 2.2 Workload Prediction and Dynamic Provisioning . . . 10
- 2.3 Autoscaling and Reinforcement Learning . . . 10
- 2.4 Reviews of Learning-Based Resource Scheduling . . . 11
- 2.5 Workload Traces and Evaluation . . . 12
- 2.6 Research Gap . . . 12

3 Requirement Analysis **14**
- 3.1 Description and Functions of the Product . . . 14
  - 3.1.1 Product Perspective . . . 14
  - 3.1.2 Product Functions . . . 14
  - 3.1.3 Actors . . . 15
- 3.2 Specific Requirements . . . 16
  - 3.2.1 Functional Requirements . . . 16
  - 3.2.2 Non-Functional Requirements . . . 17
  - 3.2.3 System Requirements . . . 18
  - 3.2.4 Design Constraints . . . 19
  - 3.2.5 Acceptance Criteria . . . 20
- 3.3 Future Scope . . . 20

4 Algorithm Analysis and Mathematical Modeling **21**
- 4.1 System Overview . . . 21
- 4.2 Workload Data Preprocessing . . . 22
  - 4.2.1 Workload Representation . . . 22
  - 4.2.2 Data Cleaning and Normalisation . . . 22
  - 4.2.3 Time-Series Windowing . . . 22
- 4.3 CPU and Memory Demand Prediction . . . 23
  - 4.3.1 LSTM Model . . . 23
  - 4.3.2 BiLSTM Model . . . 24
  - 4.3.3 Prediction Evaluation . . . 24
- 4.4 Resource Optimization Models . . . 25

4.4.1 Particle Swarm Optimization (PSO) . . . 25
4.4.2 Ant Colony Optimization (ACO) . . . 26
4.5 SLA-Constrained Minimum-Resource Quantification . . . 27
4.5.1 Problem Formulation . . . 27
4.5.2 Candidate Configuration Search . . . 28
4.6 Reinforcement-Learning-Based Proactive Allocation . . . 29
4.6.1 State Representation . . . 29
4.6.2 Action Space . . . 29
4.6.3 Reward Function . . . 30
4.6.4 Learning Process . . . 30
4.7 CloudSim Plus Simulation and Evaluation . . . 31
4.8 Summary . . . 32

5 Detailed Design **33**
5.1 System Architecture . . . 33
5.2 UML Diagrams . . . 34
5.3 System Workflow . . . 36

6 Project Planning **38**

7 Results and Implementation **40**
7.1 Implementation Screenshots . . . 40

8 Conclusion **44**

List of Figures

5.1 UML Diagram 1 34
5.2 UML Diagram 2 35
5.3 UML Diagram 3 35
5.4 UML Diagram 4 35
5.5 UML Diagram 5 36

7.1 Code Workspace Creation (Reference 1.jpeg) 40
7.2 Resource Optimizer Dashboard (Reference 2.jpeg) 40
7.3 IDE Environment (Reference 3.jpeg) 41
7.4 Machine Learning Optimizer View (Reference 4.jpeg) 41
7.5 Infrastructure Launch View (Reference 5.jpeg) 41
7.6 Room Management (Reference 6.jpeg) 42
7.7 Login Interface (Reference 7.jpeg) 42

Chapter 1

Introduction

1.1 Details of project work

In cloud platforms, CPU and memory resources need to be dynamically assigned based on the current workload. Assigning too many resources would cause wastage which increases cost. Assigning too few resources can cause performance issues in the applications. Traditional platforms often assign resources in a reactive manner.

Using machine learning techniques we can forecast future workloads. However, this does not tell us how much CPU and memory is required to keep the application within its SLAs. We have created a system to forecast workloads and estimate the minimum amount of CPU and memory resources needed to meet certain performance requirements. This can be used to dynamically assign resources, through the use of reinforcement learning. The system was developed using the CloudSim Plus simulator.

This is an important issue for cloud providers, platform administrators and tenants as their applications rely on a certain service level agreement. The problem requires knowledge in multiple areas of computer engineering such as time series prediction, resource optimisation, reinforcement learning and simulation.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante.

Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

1.2 Objectives

The main objective is to estimate the minimum CPU and memory resources needed for predicted workloads while meeting SLA requirements and avoiding unnecessary allocation. The project has the following specific objectives:

1. Estimate the minimum CPU and memory resources required for predicted workloads.

2. Meet the required performance levels while reducing SLA violations.

3. Reduce unnecessary resource allocation and over-provisioning.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia

nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo. Aliquam vestibulum fringilla lorem. Sed neque lectus, consectetuer at, consectetuer sed, eleifend ac, lectus. Nulla facilisi. Pellentesque eget lectus. Proin eu metus. Sed porttitor. In hac habitasse platea dictumst. Suspendisse eu lectus. Ut mi mi, lacinia sit amet, placerat et, mollis vitae, dui. Sed ante tellus, tristique ut, iaculis eu, malesuada ac, dui. Mauris nibh leo, facilisis non, adipiscing quis, ultrices a, dui.

1.3 Scope Of The Project

The targeted scope of this project includes:

- Workload demand prediction from historical time-series data.
- Quantification of minimum CPU and memory (RAM) required.
- Estimation of minimum resource levels that meet defined SLA requirements.
- Evaluate and optimise cost of resource allocation.
- Simulate resource allocation using reinforcement learning, guided by those estimates.

The project does not cover other computing resources such as GPU, network bandwidth or storage. Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetuer. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

1.4 Motivation of the project

Cloud workloads change over time, so providers must balance service quality with efficient resource use. Predictive models and reinforcement learning can help manage resources,

but they do not always answer a practical question directly: given an expected workload and an SLA target, what is the smallest CPU and memory allocation that can meet the target?

Simply allocating resources based on the forecast may leave an unnecessary safety margin. Reducing allocations too aggressively, however, can cause latency to rise or memory to run out. Reinforcement learning systems that explore a broad range of allocation choices may also take time to learn, and their early decisions can lead to SLA violations.

This project adds an explicit resource-estimation step between prediction and allocation:

$$\text{Predicted workload} \rightarrow \text{minimum SLA-compliant CPU and memory guided} \rightarrow \text{reinforcement-learning allocation}$$

By estimating CPU and memory together, the system aims to make resource decisions more practical and easier to interpret. It turns workload forecasts into clear resource targets that can guide allocation. Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetuer tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Chapter 2

Literature Survey

Cloud resource management must balance application performance with efficient use of CPU and memory. Static allocation can waste resources when demand is low, while reactive scaling may respond too late when demand rises. Previous work has explored workload prediction, dynamic provisioning, autoscaling, and reinforcement learning to address these challenges.

2.1 Production Cloud Resource Management

Verma et al. describe Borg, Google's cluster management system, which schedules and monitors jobs across large clusters. Borg uses techniques such as task packing, admission control, over-commitment, and machine sharing to improve resource utilisation while supporting application reliability. Cortez et al. present Resource Central, which uses historical virtual-machine telemetry to predict future resource behaviour. Their work shows how predictions can help cloud resource managers increase server utilisation while avoiding physical resource exhaustion. Rzadca et al. describe Autopilot, which uses historical workload data, machine-learning models, and heuristics to recommend CPU and memory limits and adjust the number of concurrent tasks. Its goal is to reduce unused capacity while limiting CPU throttling and out-of-memory failures. Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor.

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo. Sed felis erat, ullamcorper in, dictum non, ultricies ut, lectus. Proin vel arcu a odio lobortis euismod. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin ut est. Aliquam odio. Pellentesque massa turpis, cursus eu, euismod nec, tempor congue, nulla. Duis viverra gravida mauris. Cras tincidunt. Curabitur eros ligula, varius ut, pulvinar in,

cursus faucibus, augue.

Nulla mattis luctus nulla. Duis commodo velit at leo. Aliquam vulputate magna et leo. Nam vestibulum ullamcorper leo. Vestibulum condimentum rutrum mauris. Donec id mauris. Morbi molestie justo et pede. Vivamus eget turpis sed nisl cursus tempor. Curabitur mollis sapien condimentum nunc. In wisi nisl, malesuada at, dignissim sit amet, lobortis in, odio. Aenean consequat arcu a ante. Pellentesque porta elit sit amet orci. Etiam at turpis nec elit ultricies imperdiet. Nulla facilisi. In hac habitasse platea dictumst. Suspendisse viverra aliquam risus. Nullam pede justo, molestie nonummy, scelerisque eu, facilisis vel, arcu.

2.2 Workload Prediction and Dynamic Provisioning

Panwar and Supriya propose a dynamic resource-provisioning approach that combines linear regression, Bayesian learning, and autonomic computing. They evaluate it in CloudSim using Google and ClarkNet workload traces, measuring response time, SLA violations, VM usage, and cost. Wang and Yang combine LSTM-based workload prediction with a Deep Q-Network for dynamic resource scheduling. Kamma et al. similarly combine workload forecasting with reinforcement-learning-based resource allocation. These studies illustrate how forecasts can inform allocation decisions. The present project builds on this idea by adding an explicit estimate of the minimum CPU and memory required to meet an SLA target. Curabitur tellus magna, porttitor a, commodo a, commodo in, tortor. Donec interdum. Praesent scelerisque. Maecenas posuere sodales odio. Vivamus metus lacus, varius quis, imperdiet quis, rhoncus a, turpis. Etiam ligula arcu, elementum a, venenatis quis, sollicitudin sed, metus. Donec nunc pede, tincidunt in, venenatis vitae, faucibus vel, nibh. Pellentesque wisi. Nullam malesuada. Morbi ut tellus ut pede tincidunt porta. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam congue neque id dolor.

Donec et nisl at wisi luctus bibendum. Nam interdum tellus ac libero. Sed sem justo, laoreet vitae, fringilla at, adipiscing ut, nibh. Maecenas non sem quis tortor eleifend fermentum. Etiam id tortor ac mauris porta vulputate. Integer porta neque vitae massa. Maecenas tempus libero a libero posuere dictum. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Aenean quis mauris sed elit commodo placerat. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Vivamus rhoncus tincidunt libero. Etiam elementum pretium justo. Vivamus est. Morbi a tellus eget pede tristique commodo. Nulla nisl. Vestibulum sed nisl eu sapien cursus rutrum.

2.3 Autoscaling and Reinforcement Learning

Mishra et al. explore reinforcement-learning-based autoscaling as an alternative to fixed scaling rules. Their approach adjusts decisions as workload conditions change. Sharma and Patel compare several scheduling methods, including reinforcement learning, using Alibaba cluster workload data in simulation. Their results support the use of adaptive scheduling for improving job completion time and reducing slowdown. Kayalvili et al. propose a prediction-enabled resource-allocation framework that combines Q-learning with machine-learning models to estimate Q-values and guide allocation decisions. They evaluate their approach using CloudStack and RUBIS benchmarks, reporting reductions

in SLA violations and resource costs compared with round-robin scheduling. Narsimhulu and Kumar combine reinforcement learning, a genetic algorithm, and an LSTM autoencoder for cloud task scheduling. Their system uses the autoencoder to predict overload conditions and detect abnormal behaviour, while the other components guide task allocation. Their evaluation includes SLA violations, CPU utilisation, makespan, and energy use. Li et al. propose HiGFRL, a graph-based reinforcement-learning framework for scheduling dependency-aware tasks in heterogeneous cloud clusters. It combines task-dependency and resource-state information to guide task-to-node allocation. The paper evaluates the method using Alibaba cluster traces and reports improvements in scheduling metrics. Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra. Nullam varius. Etiam dignissim elementum metus. Vestibulum faucibus, metus sit amet mattis rhoncus, sapien dui laoreet odio, nec ultricies nibh augue a enim. Fusce in ligula. Quisque at magna et nulla commodo consequat. Proin accumsan imperdiet sem. Nunc porta. Donec feugiat mi at justo. Phasellus facilisis ipsum quis ante. In ac elit eget ipsum pharetra faucibus. Maecenas viverra nulla in massa.

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus. Aenean ac enim. In luctus. Phasellus eu quam vitae turpis viverra pellentesque. Duis feugiat felis ut enim. Phasellus pharetra, sem id porttitor sodales, magna nunc aliquet nibh, nec blandit nisl mauris at pede. Suspendisse risus risus, lobortis eget, semper at, imperdiet sit amet, quam. Quisque scelerisque dapibus nibh. Nam enim. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Nunc ut metus. Ut metus justo, auctor at, ultrices eu, sagittis ut, purus. Aliquam aliquam.

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor. Phasellus aliquet odio ac lacus tempor faucibus. Praesent sed sem. Praesent iaculis. Cras rhoncus tellus sed justo ullamcorper sagittis. Donec quis orci. Sed ut tortor quis tellus euismod tincidunt. Suspendisse congue nisl eu elit. Aliquam tortor diam, tempus id, tristique eget, sodales vel, nulla. Praesent tellus mi, condimentum sed, viverra at, consectetuer quis, lectus. In auctor vehicula orci. Sed pede sapien, euismod in, suscipit in, pharetra placerat, metus. Vivamus commodo dui non odio. Donec et felis.

2.4 Reviews of Learning-Based Resource Scheduling

Zhou et al. review deep-reinforcement-learning methods for cloud resource scheduling. They discuss the benefits of DRL for complex scheduling problems and identify challenges and future research directions. Bodra and Khairnar compare machine-learning approaches to cloud resource allocation, including deep reinforcement learning, neural networks, and multi-agent systems. Their review discusses the potential of hybrid approaches for addressing multiple resource-management objectives. Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci. Pellentesque mattis lacinia elit. Fusce molestie velit in ligula. Nullam et orci vitae nibh vulputate auctor. Aliquam eget purus. Nulla auctor wisi sed ipsum. Morbi porttitor tellus ac enim. Fusce ornare. Proin ipsum enim, tincidunt in, ornare venenatis, molestie a, augue. Donec vel pede in lacus sagittis porta. Sed hendrerit ipsum quis nisl. Suspendisse quis massa ac nibh pretium cursus. Sed sodales. Nam eu neque quis pede dignissim ornare. Maecenas eu purus ac urna tincidunt congue.

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi

purus. Nulla a est sit amet purus venenatis iaculis. Vivamus viverra purus vel magna. Donec in justo sed odio malesuada dapibus. Nunc ultrices aliquam nunc. Vivamus facilisis pellentesque velit. Nulla nunc velit, vulputate dapibus, vulputate id, mattis ac, justo. Nam mattis elit dapibus purus. Quisque enim risus, congue non, elementum ut, mattis quis, sem. Quisque elit.

2.5 Workload Traces and Evaluation

The Google Cluster Usage Traces v3 documentation describes the structure of Google workload data, including records related to machines, jobs, tasks, and resource usage. It also documents limitations such as obfuscated values and missing events, which should be considered during data preparation and evaluation. The reviewed studies use different datasets, simulators, baselines, and performance measures, so their reported results cannot be compared directly. This project uses CloudSim Plus to compare static allocation, reactive threshold-based allocation, prediction-only allocation, and prediction-guided reinforcement-learning allocation under consistent workload and SLA settings. Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem. Donec ut ante quis dolor bibendum condimentum. Etiam egestas tortor vitae lacus. Praesent cursus. Mauris bibendum pede at elit. Morbi et felis a lectus interdum facilisis. Sed suscipit gravida turpis. Nulla at lectus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Praesent nonummy luctus nibh. Proin turpis nunc, congue eu, egestas ut, fringilla at, tellus. In hac habitasse platea dictumst.

Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum. Suspendisse condimentum, tortor at egestas posuere, neque metus tempor orci, et tincidunt urna nunc a purus. Sed facilisis blandit tellus. Nunc risus sem, suscipit nec, eleifend quis, cursus quis, libero. Curabitur et dolor. Sed vitae sem. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Maecenas ante. Duis ullamcorper enim. Donec tristique enim eu leo. Nullam molestie elit eu dolor. Nullam bibendum, turpis vitae tristique gravida, quam sapien tempor lectus, quis pretium tellus purus ac quam. Nulla facilisi.

2.6 Research Gap

Previous work shows that workload prediction can support cloud resource management and that reinforcement learning can adapt scheduling and autoscaling decisions to changing demand. Production systems such as Resource Central and Autopilot also demonstrate the value of historical workload data in guiding CPU and memory recommendations. However, much of the reviewed work focuses on prediction accuracy, utilisation, cost, task scheduling, or autoscaling. This project addresses a specific operational question: given a predicted workload and an SLA target, what is the minimum CPU and memory configuration expected to meet that target? The estimated minimum resources will then guide proactive allocation through reinforcement learning. Duis aliquet dui in est. Donec eget est. Nunc lectus odio, varius at, fermentum in, accumsan non, enim. Aliquam erat volutpat. Proin sit amet nulla ut eros consectetuer cursus. Phasellus dapibus aliquam justo. Nunc laoreet. Donec consequat placerat magna. Duis pretium tincidunt

justo. Sed sollicitudin vestibulum quam. Nam quis ligula. Vivamus at metus. Etiam imperdiet imperdiet pede. Aenean turpis. Fusce augue velit, scelerisque sollicitudin, dictum vitae, tempor et, pede. Donec wisi sapien, feugiat in, fermentum ut, sollicitudin adipiscing, metus.

Donec vel nibh ut felis consectetuer laoreet. Donec pede. Sed id quam id wisi laoreet suscipit. Nulla lectus dolor, aliquam ac, fringilla eget, mollis ut, orci. In pellentesque justo in ligula. Maecenas turpis. Donec eleifend leo at felis tincidunt consequat. Aenean turpis metus, malesuada sed, condimentum sit amet, auctor a, wisi. Pellentesque sapien elit, bibendum ac, posuere et, congue eu, felis. Vestibulum mattis libero quis metus scelerisque ultrices. Sed purus.

Donec molestie, magna ut luctus ultrices, tellus arcu nonummy velit, sit amet pulvinar elit justo et mauris. In pede. Maecenas euismod elit eu erat. Aliquam augue wisi, facilisis congue, suscipit in, adipiscing et, ante. In justo. Cras lobortis neque ac ipsum. Nunc fermentum massa at ante. Donec orci tortor, egestas sit amet, ultrices eget, venenatis eget, mi. Maecenas vehicula leo semper est. Mauris vel metus. Aliquam erat volutpat. In rhoncus sapien ac tellus. Pellentesque ligula.

Chapter 3

Requirement Analysis

3.1 Description and Functions of the Product

3.1.1 Product Perspective

The proposed system is a research prototype for cloud resource management. It combines workload prediction, SLA-constrained resource quantification, reinforcement learning, and cloud simulation. Python will be used for workload preprocessing and machine-learning components. CloudSim Plus, which is Java-based, will provide the simulation environment. The Python components and CloudSim Plus will exchange workload, prediction, allocation, and simulation information through a defined interface. A file-based interface such as CSV or a lightweight API may be used, depending on the final implementation. Cras dapibus, augue quis scelerisque ultricies, felis dolor placerat sem, id porta velit odio eu elit. Aenean interdum nibh sed wisi. Praesent sollicitudin vulputate dui. Praesent iaculis viverra augue. Quisque in libero. Aenean gravida lorem vitae sem ullamcorper cursus. Nunc adipiscing rutrum ante. Nunc ipsum massa, faucibus sit amet, viverra vel, elementum semper, orci. Cras eros sem, vulputate et, tincidunt id, ultrices eget, magna. Nulla varius ornare odio. Donec accumsan mauris sit amet augue. Sed ligula lacus, laoreet non, aliquam sit amet, iaculis tempor, lorem. Suspendisse eros. Nam porta, leo sed congue tempor, felis est ultrices eros, id mattis velit felis non metus. Curabitur vitae elit non mauris varius pretium. Aenean lacus sem, tincidunt ut, consequat quis, porta vitae, turpis. Nullam laoreet fermentum urna. Proin iaculis lectus.

Sed mattis, erat sit amet gravida malesuada, elit augue egestas diam, tempus scelerisque nunc nisl vitae libero. Sed consequat feugiat massa. Nunc porta, eros in eleifend varius, erat leo rutrum dui, non convallis lectus orci ut nibh. Sed lorem massa, nonummy quis, egestas id, condimentum at, nisl. Maecenas at nibh. Aliquam et augue at nunc pellentesque ullamcorper. Duis nisl nibh, laoreet suscipit, convallis ut, rutrum id, enim. Phasellus odio. Nulla nulla elit, molestie non, scelerisque at, vestibulum eu, nulla. Ut odio nisl, facilisis id, mollis et, scelerisque nec, enim. Aenean sem leo, pellentesque sit amet, scelerisque sit amet, vehicula pellentesque, sapien.

3.1.2 Product Functions

The system will perform the following functions:

1. Load historical CPU and memory workload data.

2. Preprocess the data and prepare time-series sequences.
3. Predict future CPU and memory demand.
4. Evaluate candidate CPU and memory configurations against the defined SLA condition.
5. Report the minimum feasible configuration (C_{min}, M_{min}).
6. Use a reinforcement-learning controller to produce proactive allocation actions.
7. Apply the allocation policy in CloudSim Plus.
8. Record SLA compliance, resource utilisation, resource wastage, allocation cost, and execution or response time.
9. Compare the proposed method with the selected baseline allocation approaches.

Sed consequat tellus et tortor. Ut tempor laoreet quam. Nullam id wisi a libero tristique semper. Nullam nisl massa, rutrum ut, egestas semper, mollis id, leo. Nulla ac massa eu risus blandit mattis. Mauris ut nunc. In hac habitasse platea dictumst. Aliquam eget tortor. Quisque dapibus pede in erat. Nunc enim. In dui nulla, commodo at, consectetuer nec, malesuada nec, elit. Aliquam ornare tellus eu urna. Sed nec metus. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Phasellus id magna. Duis malesuada interdum arcu. Integer metus. Morbi pulvinar pellentesque mi. Suspendisse sed est eu magna molestie egestas. Quisque mi lorem, pulvinar eget, egestas quis, luctus at, ante. Proin auctor vehicula purus. Fusce ac nisl aliquam ante hendrerit pellentesque. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Morbi wisi. Etiam arcu mauris, facilisis sed, eleifend non, nonummy ut, pede. Cras ut lacus tempor metus mollis placerat. Vivamus eu tortor vel metus interdum malesuada.

3.1.3 Actors

- **Cloud Service Provider:** Interested in resource utilisation, allocation cost, and SLA compliance.
- **Cloud Customer or Application Owner:** Provides workload information and specifies or agrees to the SLA target.
- **Cloud Administrator:** Reviews resource-allocation decisions and simulation performance.
- **Project Researcher or Developer:** Configures experiments, runs the simulation, and analyses results.
- **End User:** Does not operate the system directly but is affected by the response time and availability of the simulated application.

Sed eleifend, eros sit amet faucibus elementum, urna sapien consectetuer mauris, quis egestas leo justo non risus. Morbi non felis ac libero vulputate fringilla. Mauris libero eros, lacinia non, sodales quis, dapibus porttitor, pede. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Morbi dapibus mauris condimentum nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Etiam sit amet erat. Nulla varius. Etiam tincidunt dui vitae turpis. Donec leo. Morbi vulputate convallis est. Integer aliquet. Pellentesque aliquet sodales urna.

3.2 Specific Requirements

3.2.1 Functional Requirements

1. **FR1 - Workload data ingestion:** The system shall accept historical workload data containing CPU and memory measurements.

2. **FR2 - Data preprocessing:** The system shall clean, normalise, and transform workload data into time-series sequences suitable for prediction.

3. **FR3 - CPU demand prediction:** The system shall predict future CPU demand from historical workload data.

4. **FR4 - Memory demand prediction:** The system shall predict future memory demand from historical workload data.

5. **FR5 - Minimum-resource quantification:** The system shall determine a minimum feasible CPU and memory configuration for a predicted workload under the defined SLA constraint.

6. **FR6 - SLA evaluation:** The system shall evaluate whether a candidate CPU and memory configuration satisfies the defined SLA condition.

7. **FR7 - Reinforcement-learning allocation:** The system shall use a reinforcement-learning controller to select proactive CPU and memory allocation actions, guided by the quantified minimum-resource configuration.

8. **FR8 - Cloud simulation:** The system shall execute the selected allocation policy in CloudSim Plus.

9. **FR9 - Performance monitoring:** The system shall record CPU utilisation, memory utilisation, SLA violations, resource wastage, allocation cost, and execution or response time.

10. **FR10 - Baseline comparison:** The system shall compare the proposed method with static allocation, reactive threshold-based allocation, and prediction-only allocation.

11. **FR11 - Component evaluation:** The system shall support an ablation experiment to measure the effect of the minimum-resource quantification stage.

Nullam eleifend justo in nisl. In hac habitasse platea dictumst. Morbi nonummy. Aliquam ut felis. In velit leo, dictum vitae, posuere id, vulputate nec, ante. Maecenas vitae pede nec dui dignissim suscipit. Morbi magna. Vestibulum id purus eget velit laoreet laoreet. Praesent sed leo vel nibh convallis blandit. Ut rutrum. Donec nibh. Donec interdum. Fusce sed pede sit amet elit rhoncus ultrices. Nullam at enim vitae pede vehicula iaculis.

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio. Integer euismod imperdiet turpis. Ut nec leo nec diam imperdiet lacinia. Etiam eget lacus eget mi ultricies posuere. In placerat tristique tortor. Sed porta vestibulum metus. Nulla iaculis sollicitudin pede. Fusce luctus tellus in dolor. Curabitur auctor velit a sem. Morbi sapien. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec adipiscing urna vehicula nunc. Sed ornare leo in leo. In rhoncus leo ut dui. Aenean dolor quam, volutpat nec, fringilla id, consectetuer vel, pede.

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In hac habitasse platea dictumst. Nullam molestie turpis eget nisl. Duis a massa id pede dapibus ultricies. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor.

3.2.2 Non-Functional Requirements

1. **NFR1 - Prediction accuracy:** The prediction module should be evaluated using MAE and RMSE and compared with at least one simpler prediction baseline.

2. **NFR2 - SLA reliability:** The allocation policy shall be evaluated against a predefined SLA condition. The SLA formula and threshold shall be stated before the experiments; no universal violation threshold is assumed.

3. **NFR3 - Reproducibility:** Experiments shall use recorded configurations and fixed random seeds where applicable. Results should be reported across multiple runs.

4. **NFR4 - Scalability:** The framework should be evaluated with different workload sizes or trace subsets.

5. **NFR5 - Performance:** The reinforcement-learning decision process should not add unacceptable computational overhead to the simulation.

6. **NFR6 - Maintainability:** Workload preprocessing, prediction, minimum-resource quantification, reinforcement learning, and simulation should be implemented as separate modules.

7. **NFR7 - Extensibility:** The design should allow additional resource types to be considered in future work, although they are outside the current project scope.

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim. Nullam congue suscipit nibh. Quisque convallis. Praesent arcu nibh, vehicula eget, accumsan eu, tincidunt a, nibh. Suspendisse vulputate, tortor quis adipiscing viverra, lacus nibh dignissim tellus, eu suscipit risus ante fringilla diam. Quisque a libero vel pede imperdiet aliquet. Pellentesque nunc nibh, eleifend a, consequat consequat, hendrerit nec, diam. Sed urna. Maecenas laoreet eleifend neque. Vivamus purus odio, eleifend non, iaculis a, ultrices sit amet, urna. Mauris faucibus odio vitae risus. In nisl. Praesent purus. Integer iaculis, sem eu egestas lacinia, lacus pede scelerisque augue, in ullamcorper dolor eros ac lacus. Nunc in libero.

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo. Aenean tincidunt. Donec suscipit. Cras id justo quis nibh scelerisque dignissim. Aliquam sagittis elementum dolor. Aenean consectetuer justo in pede. Curabitur ullamcorper ligula nec orci. Aliquam purus turpis, aliquam id, ornare vitae, porttitor non, wisi. Maecenas luctus porta lorem. Donec vitae ligula eu ante pretium varius. Proin tortor metus, convallis et, hendrerit non, scelerisque in, urna. Cras quis libero eu ligula bibendum tempor. Vivamus tellus quam, malesuada eu, tempus sed, tempor sed, velit. Donec lacinia auctor libero.

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus. Ut quis nunc sed odio malesuada egestas. Duis a magna sit amet ligula tristique pretium. Ut pharetra. Vestibulum imperdiet magna nec wisi. Mauris convallis. Sed accumsan sollicitudin massa. Sed id enim. Nunc pede enim, lacinia ut, pulvinar quis, suscipit semper, elit. Cras accumsan erat vitae enim. Cras sollicitudin. Vestibulum rutrum blandit massa.

3.2.3 System Requirements

1. **Dataset requirement:** The primary dataset shall be the Google Cluster Usage Trace. A secondary dataset may be used for validation if it is selected and made available within the project schedule.

2. **Hardware requirements:**

   - A standard development computer capable of running Python and Java.
   - Sufficient memory and storage to process the selected workload trace and run CloudSim Plus experiments.
   - A dedicated GPU is not required for the core proof-of-concept, although it may speed up model training.

3. **Software requirements:**

   - Python 3 for preprocessing, prediction, and reinforcement-learning components.
   - PyTorch or TensorFlow for LSTM or BiLSTM models.
   - Pandas and NumPy for data processing.
   - Scikit-learn for baseline models and evaluation metrics.
   - Java and CloudSim Plus for cloud simulation.

- Git for version control and experiment tracking.

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis. Integer sollicitudin augue nec dui. Fusce lectus. Vivamus faucibus nulla nec lacus. Integer diam. Pellentesque sodales, enim feugiat cursus volutpat, sem mauris dignissim mauris, quis consequat sem est fermentum ligula. Nullam justo lectus, condimentum sit amet, posuere a, fringilla mollis, felis. Morbi nulla nibh, pellentesque at, nonummy eu, sollicitudin nec, ipsum. Cras neque. Nunc augue. Nullam vitae quam id quam pulvinar blandit. Nunc sit amet orci. Aliquam erat elit, pharetra nec, aliquet a, gravida in, mi. Quisque urna enim, viverra quis, suscipit quis, tincidunt ut, sapien. Cras placerat consequat sem. Curabitur ac diam. Curabitur diam tortor, mollis et, viverra ac, tempus vel, metus.

Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit. Praesent convallis sapien ac est. Aliquam ullamcorper euismod nulla. Integer mollis enim vel tortor. Nulla sodales placerat nunc. Sed tempus rutrum wisi. Duis accumsan gravida purus. Nunc nunc. Etiam facilisis dui eu sem. Vestibulum semper. Praesent eu eros. Vestibulum tellus nisl, dapibus id, vestibulum sit amet, placerat ac, mauris. Maecenas et elit ut erat placerat dictum. Nam feugiat, turpis et sodales volutpat, wisi quam rhoncus neque, vitae aliquam ipsum sapien vel enim. Maecenas suscipit cursus mi.

3.2.4 Design Constraints

- The system shall model and quantify CPU and memory only.
- All allocation experiments shall be conducted in CloudSim Plus; deployment on a commercial cloud is outside the project scope.
- The SLA condition and its threshold shall be defined as part of the experimental design.
- The minimum-resource search shall initially use discrete candidate CPU and memory configurations to keep the search bounded and auditable.
- The reinforcement-learning action space shall be limited to a manageable set of CPU and memory allocation choices.
- The system shall not claim improved performance unless the results support that conclusion in the specified experiments.

Quisque consectetuer. In suscipit mauris a dolor pellentesque consectetuer. Mauris convallis neque non erat. In lacinia. Pellentesque leo eros, sagittis quis, fermentum quis, tincidunt ut, sapien. Maecenas sem. Curabitur eros odio, interdum eu, feugiat eu, porta ac, nisl. Curabitur nunc. Etiam fermentum convallis velit. Pellentesque laoreet lacus. Quisque sed elit. Nam quis tellus. Aliquam tellus arcu, adipiscing non, tincidunt eleifend, adipiscing quis, augue. Vivamus elementum placerat enim. Suspendisse ut tortor. Integer faucibus adipiscing felis. Aenean consectetuer mattis lectus. Morbi malesuada faucibus dolor. Nam lacus. Etiam arcu libero, malesuada vitae, aliquam vitae, blandit tristique, nisl.

Maecenas accumsan dapibus sapien. Duis pretium iaculis arcu. Curabitur ut lacus. Aliquam vulputate. Suspendisse ut purus sed sem tempor rhoncus. Ut quam dui, fringilla

at, dictum eget, ultricies quis, quam. Etiam sem est, pharetra non, vulputate in, pretium at, ipsum. Nunc semper sagittis orci. Sed scelerisque suscipit diam. Ut volutpat, dolor at ullamcorper tristique, eros purus mollis quam, sit amet ornare ante nunc et enim.

3.2.5 Acceptance Criteria

1. The system successfully loads the selected workload data.
2. Preprocessing produces valid CPU and memory sequences with missing values handled.
3. The prediction module produces both CPU and memory forecasts.
4. Prediction performance is measured using MAE and RMSE and compared with at least one baseline predictor.
5. The quantification module produces (C_{min}, M_{min}) for a specified SLA condition.
6. Candidate allocations are classified as SLA-compliant or non-compliant using the defined SLA formulation.
7. The reinforcement-learning controller produces CPU and memory allocation actions.
8. The proposed allocation policy runs in CloudSim Plus.
9. The system records SLA violations, utilisation, wastage, cost, and execution or response time.
10. The proposed approach is compared with at least three baselines.
11. Results are repeated across multiple runs or seeds where applicable.
12. The evaluation determines whether the proposed method reduces unnecessary allocation while meeting the specified SLA condition.

3.3 Future Scope

Future work may evaluate the system using additional workload datasets, extend resource quantification to bandwidth or storage, or study deployment in a container-orchestration environment. These extensions are outside the present project scope.

Chapter 4

Algorithm Analysis and Mathematical Modeling

4.1 System Overview

The proposed system has four main stages: workload preprocessing, CPU and memory prediction, SLA-constrained minimum-resource quantification, and proactive allocation using reinforcement learning. The allocation policy is evaluated in CloudSim Plus. The overall process is:

Historical workload → preprocessing → CPU and memory prediction → minimum-resource quantification → RL-based allocation → evaluation in CloudSim Plus

At each prediction interval, the system uses historical workload observations to estimate future CPU and memory demand. The quantification module then searches for a feasible resource configuration that meets the defined SLA condition. The RL controller uses this configuration, along with the current simulation state, to select an allocation action. Simulation results provide feedback for evaluating the allocation policy. Phasellus fringilla, metus id feugiat consectetuer, lacus wisi ultrices tellus, quis lobortis nibh lorem quis tortor. Donec egestas ornare nulla. Mauris mi tellus, porta faucibus, dictum vel, nonummy in, est. Aliquam erat volutpat. In tellus magna, porttitor lacinia, molestie vitae, pellentesque eu, justo. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed orci nibh, scelerisque sit amet, suscipit sed, placerat vel, diam. Vestibulum nonummy vulputate orci. Donec et velit ac arcu interdum semper. Morbi pede orci, cursus ac, elementum non, vehicula ut, lacus. Cras volutpat. Nam vel wisi quis libero venenatis placerat. Aenean sed odio. Quisque posuere purus ac orci. Vivamus odio. Vivamus varius, nulla sit amet semper viverra, odio mauris consequat lacus, at vestibulum neque arcu eu tortor. Donec iaculis tincidunt tellus. Aliquam erat volutpat. Curabitur magna lorem, dignissim volutpat, viverra et, adipiscing nec, dolor. Praesent lacus mauris, dapibus vitae, sollicitudin sit amet, nonummy eget, ligula.

Cras egestas ipsum a nisl. Vivamus varius dolor ut dolor. Fusce vel enim. Pellentesque accumsan ligula et eros. Cras id lacus non tortor facilisis facilisis. Etiam nisl elit, cursus sed, fringilla in, congue nec, urna. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer at turpis. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Duis fringilla, ligula sed porta fringilla, ligula wisi commodo felis, ut adipiscing felis dui in enim. Suspendisse malesuada

ultrices ante. Pellentesque scelerisque augue sit amet urna. Nulla volutpat aliquet tortor. Cras aliquam, tellus at aliquet pellentesque, justo sapien commodo leo, id rhoncus sapien quam at erat. Nulla commodo, wisi eget sollicitudin pretium, orci orci aliquam orci, ut cursus turpis justo et lacus. Nulla vel tortor. Quisque erat elit, viverra sit amet, sagittis eget, porta sit amet, lacus.

In hac habitasse platea dictumst. Proin at est. Curabitur tempus vulputate elit. Pellentesque sem. Praesent eu sapien. Duis elit magna, aliquet at, tempus sed, vehicula non, enim. Morbi viverra arcu nec purus. Vivamus fringilla, enim et commodo malesuada, tortor metus elementum ligula, nec aliquet est sapien ut lectus. Aliquam mi. Ut nec elit. Fusce euismod luctus tellus. Curabitur scelerisque. Nullam purus. Nam ultricies accumsan magna. Morbi pulvinar lorem sit amet ipsum. Donec ut justo vitae nibh mollis congue. Fusce quis diam. Praesent tempus eros ut quam.

4.2 Workload Data Preprocessing

4.2.1 Workload Representation

Let the workload at time step t be represented by:

$$x_t = \begin{bmatrix} x_t^{(CPU)} \\ x_t^{(MEM)} \end{bmatrix} \tag{4.1}$$

where $x_t^{(CPU)}$ and $x_t^{(MEM)}$ are the observed CPU and memory demand, respectively. The workload trace is divided into time-ordered observations. Each observation is associated with a time step so the prediction model can learn changes in demand over time. The selected trace and its available fields determine the time interval and units used in the experiments.

4.2.2 Data Cleaning and Normalisation

Missing or invalid measurements shall be identified during preprocessing. Any removal or imputation of values shall be applied consistently and recorded as part of the experiment configuration. To place CPU and memory values on comparable scales, each feature may be normalised using min-max scaling:

$$x'_t = \frac{x_t - x_{min}}{x_{max} - x_{min}} \tag{4.2}$$

where x_{min} and x_{max} are calculated from the training data. The same training-set values shall be used to transform validation and test data to avoid information leakage. If the model output is normalised, predicted values shall be transformed back to the original resource units before minimum-resource quantification.

4.2.3 Time-Series Windowing

A sliding window of length L is used to form an input sequence:

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t] \tag{4.3}$$

The model uses this sequence to predict CPU and memory demand for a future horizon H:

$$\hat{Y}_t = [\hat{x}_{t+1}, \hat{x}_{t+2}, \ldots, \hat{x}_{t+H}] \tag{4.4}$$

The values of L and H shall be selected and reported as part of the experimental setup. Donec in nisl. Fusce vitae est. Vivamus ante ante, mattis laoreet, posuere eget, congue vel, nunc. Fusce sem. Nam vel orci eu eros viverra luctus. Pellentesque sit amet augue. Nunc sit amet ipsum et lacus varius nonummy. Integer rutrum sem eget wisi. Aenean eu sapien. Quisque ornare dignissim mi. Duis a urna vel risus pharetra imperdiet. Suspendisse potenti.

Morbi justo. Aenean nec dolor. In hac habitasse platea dictumst. Proin nonummy porttitor velit. Sed sit amet leo nec metus rhoncus varius. Cras ante. Vestibulum commodo sem tincidunt massa. Nam justo. Aenean luctus, felis et condimentum lacinia, lectus enim pulvinar purus, non porta velit nisl sed eros. Suspendisse consequat. Mauris a dui et tortor mattis pretium. Sed nulla metus, volutpat id, aliquam eget, ullamcorper ut, ipsum. Morbi eu nunc. Praesent pretium. Duis aliquam pulvinar ligula. Ut blandit egestas justo. Quisque posuere metus viverra pede.

Vivamus sodales elementum neque. Vivamus dignissim accumsan neque. Sed at enim. Vestibulum nonummy interdum purus. Mauris ornare velit id nibh pretium ultricies. Fusce tempor pellentesque odio. Vivamus augue purus, laoreet in, scelerisque vel, commodo id, wisi. Duis enim. Nulla interdum, nunc eu semper eleifend, enim dolor pretium elit, ut commodo ligula nisl a est. Vivamus ante. Nulla leo massa, posuere nec, volutpat vitae, rhoncus eu, magna.

4.3 CPU and Memory Demand Prediction

4.3.1 LSTM Model

An LSTM model is considered for predicting future CPU and memory demand from the historical sequence. At each time step, the LSTM updates its internal state using the current input and information retained from earlier time steps. This allows it to learn temporal patterns in the workload. The LSTM cell can be represented generally as:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{4.5}$$
$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \tag{4.6}$$
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \tag{4.7}$$
$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \tag{4.8}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{4.9}$$
$$h_t = o_t \odot \tanh(c_t) \tag{4.10}$$

where i_t, f_t, and o_t are the input, forget, and output gates; c_t is the cell state; and h_t is the hidden state. A prediction layer maps the final hidden representation to the CPU and memory forecasts:

$$\hat{x}_{t+1} = W_y h_t + b_y = \begin{bmatrix} \hat{x}_{t+1}^{(CPU)} \\ \hat{x}_{t+1}^{(MEM)} \end{bmatrix} \tag{4.11}$$

4.3.2 BiLSTM Model

A BiLSTM model may also be evaluated. It processes each input window in both the forward and reverse directions and combines the resulting hidden representations:

$$h = [\vec{h}, \overleftarrow{h}] \tag{4.12}$$

The LSTM and BiLSTM models shall be compared using the same data split and evaluation metrics. The better-performing model may be selected for the downstream experiments based on validation data. Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetuer lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetuer eget, vulputate sit amet, erat.

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

4.3.3 Prediction Evaluation

Prediction performance shall be measured separately for CPU and memory. For actual values y_i and predictions $\hat{y}_i$ over N observations, the mean absolute error (MAE) is:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{4.13}$$

The root mean squared error (RMSE) is:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \tag{4.14}$$

The prediction model shall also be compared with at least one simpler baseline, such as a moving-average predictor.

4.4 Resource Optimization Models

4.4.1 Particle Swarm Optimization (PSO)

PSO is used in our system to find the optimal continuous allocation values for CPU and memory sizing. The algorithm updates the velocity and position of each "particle" (a candidate resource allocation) across iterations to find the global best configuration. The velocity update formula dictates how the particle changes its search direction based on its own past best and the swarm's overall best:

$$v_i^{t+1} = wv_i^t + c_1 r_1(pbest_i - x_i^t) + c_2 r_2(gbest - x_i^t) \tag{4.15}$$

The position update formula calculates the new CPU/memory allocation for the next iteration:

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{4.16}$$

Where:

- v_i^t is the velocity of particle i at iteration t.
- x_i^t is the current position (resource allocation) of particle i.
- w is the inertia weight, which balances global exploration and local exploitation.
- c_1 and c_2 are the cognitive and social acceleration coefficients.
- r_1 and r_2 are random numbers uniformly distributed in $[0, 1]$.
- $pbest_i$ is the personal best position found by particle i so far.
- $gbest$ is the global best position found by the entire swarm.

Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Quisque enim arcu, suscipit nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

Morbi nunc. Aliquam consectetuer varius nulla. Phasellus eros. Cras dapibus porttitor risus. Maecenas ultrices mi sed diam. Praesent gravida velit at elit vehicula porttitor. Phasellus nisl mi, sagittis ac, pulvinar id, gravida sit amet, erat. Vestibulum est. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Curabitur id sem elementum leo rutrum hendrerit. Ut at mi. Donec tincidunt faucibus massa. Sed turpis quam, sollicitudin a, hendrerit eget, pretium ut, nisl. Duis hendrerit ligula. Nunc pulvinar congue urna.

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus.

Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut pede sem, tempus ut, porttitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

4.4.2 Ant Colony Optimization (ACO)

ACO is used in your system for discrete combinatorial optimization, specifically scheduling computational tasks to the best available Kubernetes worker nodes. The state transition rule (probability formula) determines the likelihood P_{ij}^k of a task (ant k) being scheduled from current state i to node j:

$$P_{ij}^k = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum_{l \in N_i^k} [\tau_{il}]^\alpha [\eta_{il}]^\beta} \tag{4.17}$$

The pheromone update rule happens after a scheduling iteration to reinforce paths that resulted in optimal load balancing and energy efficiency:

$$\tau_{ij}^{t+1} = (1 - \rho)\tau_{ij}^t + \Delta\tau_{ij}^t \tag{4.18}$$

Where:

- τ_{ij} is the amount of pheromone on the path to node j (representing historical success of picking this node).
- η_{ij} is the heuristic information or visibility, often inversely proportional to the node's current CPU/RAM queue length or power consumption.
- α and β are parameters that control the relative weight of the pheromone trail versus heuristic visibility.
- N_i^k is the set of valid, available nodes that ant k has not yet visited or overloaded.
- ρ is the pheromone evaporation rate ($0 < \rho \leq 1$), preventing the algorithm from converging too fast on a suboptimal node.
- $\Delta\tau_{ij}^t$ is the amount of pheromone deposited by ants that successfully scheduled tasks, typically proportional to the fitness of the allocation plan.

Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

Aenean scelerisque. Fusce pretium porttitor lorem. In hac habitasse platea dictumst. Nulla sit amet nisl at sapien egestas pretium. Nunc non tellus. Vivamus aliquet. Nam adipiscing euismod dolor. Aliquam erat volutpat. Nulla ut ipsum. Quisque tincidunt auctor augue. Nunc imperdiet ipsum eget elit. Aliquam quam leo, consectetuer non,

ornare sit amet, tristique quis, felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque interdum quam sit amet mi. Pellentesque mauris dui, dictum a, adipiscing ac, fermentum sit amet, lorem.

Ut quis wisi. Praesent quis massa. Vivamus egestas risus eget lacus. Nunc tincidunt, risus quis bibendum facilisis, lorem purus rutrum neque, nec porta tortor urna quis orci. Aenean aliquet, libero semper volutpat luctus, pede erat lacinia augue, quis rutrum sem ipsum sit amet pede. Vestibulum aliquet, nibh sed iaculis sagittis, odio dolor blandit augue, eget mollis urna tellus id tellus. Aenean aliquet aliquam nunc. Nulla ultricies justo eget orci. Phasellus tristique fermentum leo. Sed massa metus, sagittis ut, semper ut, pharetra vel, erat. Aliquam quam turpis, egestas vel, elementum in, egestas sit amet, lorem. Duis convallis, wisi sit amet mollis molestie, libero mauris porta dui, vitae aliquam arcu turpis ac sem. Aliquam aliquet dapibus metus.

Vivamus commodo eros eleifend dui. Vestibulum in leo eu erat tristique mattis. Cras at elit. Cras pellentesque. Nullam id lacus sit amet libero aliquet hendrerit. Proin placerat, mi non elementum laoreet, eros elit tincidunt magna, a rhoncus sem arcu id odio. Nulla eget leo a leo egestas facilisis. Curabitur quis velit. Phasellus aliquam, tortor nec ornare rhoncus, purus urna posuere velit, et commodo risus tellus quis tellus. Vivamus leo turpis, tempus sit amet, tristique vitae, laoreet quis, odio. Proin scelerisque bibendum ipsum. Etiam nisl. Praesent vel dolor. Pellentesque vel magna. Curabitur urna. Vivamus congue urna in velit. Etiam ullamcorper elementum dui. Praesent non urna. Sed placerat quam non mi. Pellentesque diam magna, ultricies eget, ultrices placerat, adipiscing rutrum, sem.

Morbi sem. Nulla facilisi. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nulla facilisi. Morbi sagittis ultrices libero. Praesent eu ligula sed sapien auctor sagittis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec vel nunc. Nunc fermentum, lacus id aliquam porta, dui tortor euismod eros, vel molestie ipsum purus eu lacus. Vivamus pede arcu, euismod ac, tempus id, pretium et, lacus. Curabitur sodales dapibus urna. Nunc eu sapien. Donec eget nunc a pede dictum pretium. Proin mauris. Vivamus luctus libero vel nibh.

4.5 SLA-Constrained Minimum-Resource Quantification

4.5.1 Problem Formulation

Let $\hat{D}_{CPU}$ and $\hat{D}_{MEM}$ represent predicted CPU and memory demand for the selected future interval. Let C and M represent a candidate CPU and memory allocation. A candidate configuration is feasible when it satisfies the project's defined SLA condition:

$$SLA(C, M|\hat{D}_{CPU}, \hat{D}_{MEM}) \leq \theta \tag{4.19}$$

where θ is the SLA threshold selected for the experiment. The SLA function may use the performance measures chosen for the study, such as response time or a violation indicator. Its exact definition and threshold shall be stated before running the comparative experiments. Because CPU and memory can trade off against each other, the minimum pair shall be selected using an explicit resource objective. One possible formulation is:

$$(C_{min}, M_{min}) = \arg \min_{(C,M)\in A} (w_C C + w_M M) \tag{4.20}$$

subject to:

$$SLA(C, M|\hat{D}_{CPU}, \hat{D}_{MEM}) \leq \theta \tag{4.21}$$

where A is the set of candidate allocations and w_C and w_M are the relative CPU and memory cost weights. The weights shall be specified for the experiment. If measured allocation prices are available in the simulation, those prices may be used instead of abstract weights. This formulation selects the least-cost feasible pair from the candidate set. It avoids treating the CPU minimum and memory minimum as independent values when their combination may fail to meet the SLA. Fusce tristique risus id wisi. Integer molestie massa id sem. Vestibulum vel dolor. Pellentesque vel urna vel risus ultricies elementum. Quisque sapien urna, blandit nec, iaculis ac, viverra in, odio. In hac habitasse platea dictumst. Morbi neque lacus, convallis vitae, commodo ac, fermentum eu, velit. Sed in orci. In fringilla turpis non arcu. Donec in ante. Phasellus tempor feugiat velit. Aenean varius massa non turpis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

Aliquam tortor. Morbi ipsum massa, imperdiet non, consectetuer vel, feugiat vel, lorem. Quisque eget lorem nec elit malesuada vestibulum. Quisque sollicitudin ipsum vel sem. Nulla enim. Proin nonummy felis vitae felis. Nullam pellentesque. Duis rutrum feugiat felis. Mauris vel pede sed libero tincidunt mollis. Phasellus sed urna rhoncus diam euismod bibendum. Phasellus sed nisl. Integer condimentum justo id orci iaculis varius. Quisque et lacus. Phasellus elementum, justo at dignissim auctor, wisi odio lobortis arcu, sed sollicitudin felis felis eu neque. Praesent at lacus.

Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

Etiam vel ipsum. Morbi facilisis vestibulum nisl. Praesent cursus laoreet felis. Integer adipiscing pretium orci. Nulla facilisi. Quisque posuere bibendum purus. Nulla quam mauris, cursus eget, convallis ac, molestie non, enim. Aliquam congue. Quisque sagittis nonummy sapien. Proin molestie sem vitae urna. Maecenas lorem. Vivamus viverra consequat enim.

4.5.2 Candidate Configuration Search

For the initial implementation, CPU and memory allocations shall be represented as discrete candidate values. The quantification module evaluates candidate pairs as follows:

1. Generate the set of candidate CPU and memory allocations.
2. Evaluate each pair against the defined SLA condition for the predicted workload.
3. Exclude all pairs that do not meet the SLA condition.
4. Calculate the resource objective for each remaining pair.
5. Select the feasible pair with the lowest objective value.

6. Return (C_{min}, M_{min}) and the corresponding SLA evaluation.

If no candidate pair satisfies the SLA, the module shall report that no feasible configuration was found within the candidate set. The candidate range can then be reviewed as part of the experiment setup. For n_C CPU candidates and n_M memory candidates, exhaustive evaluation checks $n_C n_M$ pairs. This bounded search is suitable for an initial implementation and produces an auditable result for each predicted workload.

4.6 Reinforcement-Learning-Based Proactive Allocation

4.6.1 State Representation

At time step t, the RL controller observes a state that may include:

$$s_t = [\hat{D}_{CPU}, \hat{D}_{MEM}, C_{min}, M_{min}, C_t, M_t, u_t^{(CPU)}, u_t^{(MEM)}, q_t] \tag{4.22}$$

where C_t and M_t are the current allocations, $u_t^{(CPU)}$ and $u_t^{(MEM)}$ are the observed utilisation values, and q_t represents the current SLA status or relevant performance measurements.

4.6.2 Action Space

The action space contains a bounded set of CPU and memory allocation changes. An action may increase, decrease, or retain the current allocation for each resource. The action set shall be designed so that all resulting configurations remain within the simulation's defined capacity limits. The controller shall use the quantified minimum resources as guidance when selecting an action. The final allocation may include a safety margin if that margin is explicitly defined and evaluated. Nunc sed pede. Praesent vitae lectus. Praesent neque justo, vehicula eget, interdum id, facilisis et, nibh. Phasellus at purus et libero lacinia dictum. Fusce aliquet. Nulla eu ante placerat leo semper dictum. Mauris metus. Curabitur lobortis. Curabitur sollicitudin hendrerit nunc. Donec ultrices lacus id ipsum.

Donec a nibh ut elit vestibulum tristique. Integer at pede. Cras volutpat varius magna. Phasellus eu wisi. Praesent risus justo, lobortis eget, scelerisque ac, aliquet in, dolor. Proin id leo. Nunc iaculis, mi vitae accumsan commodo, neque sem lacinia nulla, quis vestibulum justo sem in eros. Quisque sed massa. Morbi lectus ipsum, vulputate a, mollis ut, accumsan placerat, tellus. Nullam in wisi. Vivamus eu ligula a nunc accumsan congue. Suspendisse ac libero. Aliquam erat volutpat. Donec augue. Nunc venenatis fringilla nibh. Fusce accumsan pulvinar justo. Nullam semper, dui ut dignissim auctor, orci libero fringilla massa, blandit pulvinar pede tortor id magna. Nunc adipiscing justo sed velit tincidunt fermentum.

Integer placerat. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Sed in massa. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Phasellus tempus aliquam risus. Aliquam rutrum purus at metus. Donec posuere odio at erat. Nam non nibh. Phasellus ligula. Quisque venenatis lectus in augue. Sed vestibulum dapibus neque.

Mauris tempus eros at nulla. Sed quis dui dignissim mauris pretium tincidunt. Mauris ac purus. Phasellus ac libero. Etiam dapibus iaculis nunc. In lectus wisi, elementum

eu, sollicitudin nec, imperdiet quis, dui. Nulla viverra neque ac libero. Mauris urna leo, adipiscing eu, ultrices non, blandit eu, dui. Maecenas dui neque, suscipit sit amet, rutrum a, laoreet in, eros. Ut eu nibh. Fusce nec erat tempus urna fringilla tempus. Curabitur id enim. Sed ante. Cras sodales enim sit amet wisi. Nunc fermentum consequat quam.

Ut auctor, augue porta dignissim vestibulum, arcu diam lobortis velit, vel scelerisque risus augue sagittis risus. Maecenas eu justo. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris congue ligula eget tortor. Nullam laoreet urna sed enim. Donec eget eros ut eros volutpat convallis. Praesent turpis. Integer mauris diam, elementum quis, egestas ac, rutrum vel, orci. Nulla facilisi. Quisque adipiscing, nulla vitae elementum porta, sem urna volutpat leo, sed porta enim risus sed massa. Integer ac enim quis diam sodales luctus. Ut eget eros a ligula commodo ultricies. Donec eu urna viverra dolor hendrerit feugiat. Aliquam ac orci vel eros congue pharetra. Quisque rhoncus, justo eu volutpat faucibus, augue leo posuere lacus, a rhoncus purus pede vel est. Proin ultrices enim.

4.6.3 Reward Function

The reward shall encourage SLA compliance and discourage unnecessary resource allocation. A general reward function is:

$$r_t = -\alpha V_t - \beta K_t - \gamma W_t \tag{4.23}$$

where:

- V_t is the SLA violation penalty at time t,
- K_t is the resource allocation cost,
- W_t is the resource wastage measure, and
- α, β, and γ are reward weights.

The weights shall be selected and reported as part of the experimental configuration. The violation penalty should make SLA failures sufficiently costly for the agent to learn to avoid them, while the cost and wastage terms encourage efficient allocation.

4.6.4 Learning Process

At each simulation step, the RL controller receives the current state, selects an allocation action, and applies that action in CloudSim Plus. The simulator returns updated utilisation, performance, cost, and SLA information. These observations are used to calculate the reward and update the controller's policy. DQN or PPO may be used as the learning algorithm. The chosen algorithm, state representation, action space, reward weights, training configuration, and random seeds shall be recorded so the experiments can be reproduced. Aenean tincidunt laoreet dui. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Integer ipsum lectus, fermentum ac, malesuada in, eleifend ut, lorem. Vivamus ipsum turpis, elementum vel, hendrerit ut, semper at, metus. Vivamus sapien tortor, eleifend id, dapibus in, egestas et, pede. Pellentesque faucibus. Praesent lorem neque, dignissim in, facilisis nec, hendrerit vel, odio. Nam at diam ac neque aliquet viverra. Morbi dapibus ligula sagittis magna. In lobortis. Donec

aliquet ultricies libero. Nunc dictum vulputate purus. Morbi varius. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In tempor. Phasellus commodo porttitor magna. Curabitur vehicula odio vel dolor.

Praesent facilisis, augue a adipiscing venenatis, libero risus molestie odio, pulvinar consectetuer felis erat ac mauris. Nam vestibulum rhoncus quam. Sed velit urna, pharetra eu, eleifend eu, viverra at, wisi. Maecenas ultrices nibh at turpis. Aenean quam. Nulla ipsum. Aliquam posuere luctus erat. Curabitur magna felis, lacinia et, tristique id, ultrices ut, mauris. Suspendisse feugiat. Cras eleifend wisi vitae tortor. Phasellus leo purus, mattis sit amet, auctor in, rutrum in, magna. In hac habitasse platea dictumst. Phasellus imperdiet metus in sem. Vestibulum ac enim non sem ultricies sagittis. Sed vel diam.

Integer vel enim sed turpis adipiscing bibendum. Vestibulum pede dolor, laoreet nec, posuere in, nonummy in, sem. Donec imperdiet sapien placerat erat. Donec viverra. Aliquam eros. Nunc consequat massa id leo. Sed ullamcorper, lorem in sodales dapibus, risus metus sagittis lorem, non porttitor purus odio nec odio. Sed tincidunt posuere elit. Quisque eu enim. Donec libero risus, feugiat ac, dapibus eget, posuere a, felis. Quisque vel lectus ut metus tincidunt eleifend. Duis ut pede. Duis velit erat, venenatis vitae, vulputate a, pharetra sit amet, est. Etiam fringilla faucibus augue.

Aenean velit sem, viverra eu, tempus id, rutrum id, mi. Nullam nec nibh. Proin ullamcorper, dolor in cursus tristique, eros augue tempor nibh, at gravida diam wisi at purus. Donec mattis ullamcorper tellus. Phasellus vel nulla. Praesent interdum, eros in sodales sollicitudin, nunc nulla pulvinar justo, a euismod eros sem nec nibh. Nullam sagittis dapibus lectus. Nullam eget ipsum eu tortor lobortis sodales. Etiam purus leo, pretium nec, feugiat non, ullamcorper vel, nibh. Sed vel elit et quam accumsan facilisis. Nunc leo. Suspendisse faucibus lacus.

Pellentesque interdum sapien sed nulla. Proin tincidunt. Aliquam volutpat est vel massa. Sed dolor lacus, imperdiet non, ornare non, commodo eu, neque. Integer pretium semper justo. Proin risus. Nullam id quam. Nam neque. Duis vitae wisi ullamcorper diam congue ultricies. Quisque ligula. Mauris vehicula.

4.7 CloudSim Plus Simulation and Evaluation

The complete system shall be evaluated in CloudSim Plus. The simulation will use the selected workload trace and explicit CPU and memory utilisation models. The same workload, SLA definition, and simulation configuration shall be used when comparing the proposed method with the baselines. The evaluation shall record:

- CPU and memory prediction error.
- SLA violation rate or other selected SLA measures.
- CPU and memory utilisation.
- Resource wastage.
- Simulated allocation cost.
- Execution or response time.
- Runtime or decision overhead where applicable.

The proposed method shall be compared with at least the following baselines:

1. **Static allocation:** CPU and memory allocations remain fixed.
2. **Reactive threshold allocation:** allocation changes after observed utilisation crosses a predefined threshold.
3. **Prediction-only allocation:** predicted demand directly determines the allocation, without the minimum-resource quantification stage.
4. **Proposed method:** workload prediction followed by SLA-constrained minimum-resource quantification and RL-based proactive allocation.

An ablation experiment shall compare the proposed method with a version that omits the quantification stage. This will help determine whether the explicit minimum-resource estimate contributes to the measured outcomes.

4.8 Summary

The proposed method connects workload forecasting to resource allocation through an explicit SLA-constrained quantification stage. LSTM or BiLSTM models estimate CPU and memory demand, a bounded search identifies a minimum-cost feasible configuration, and a reinforcement-learning controller uses that configuration to guide proactive allocation in CloudSim Plus. The effectiveness of the system will be determined through reproducible comparisons with the defined baselines and evaluation measures.

Chapter 5

Detailed Design

5.1 System Architecture

The proposed system consists of workload preprocessing, CPU and memory demand prediction, SLA-constrained minimum-resource quantification, reinforcement-learning-based allocation, and CloudSim Plus simulation. Curabitur nunc magna, posuere eget, venenatis eu, vehicula ac, velit. Aenean ornare, massa a accumsan pulvinar, quam lorem laoreet purus, eu sodales magna risus molestie lorem. Nunc erat velit, hendrerit quis, malesuada ut, aliquam vitae, wisi. Sed posuere. Suspendisse ipsum arcu, scelerisque nec, aliquam eu, molestie tincidunt, justo. Phasellus iaculis. Sed posuere lorem non ipsum. Pellentesque dapibus. Suspendisse quam libero, laoreet a, tincidunt eget, consequat at, est. Nullam ut lectus non enim consequat facilisis. Mauris leo. Quisque pede ligula, auctor vel, pellentesque vel, posuere id, turpis. Cras ipsum sem, cursus et, facilisis ut, tempus euismod, quam. Suspendisse tristique dolor eu orci. Mauris mattis. Aenean semper. Vivamus tortor magna, facilisis id, varius mattis, hendrerit in, justo. Integer purus.

Vivamus adipiscing. Curabitur imperdiet tempus turpis. Vivamus sapien dolor, congue venenatis, euismod eget, porta rhoncus, magna. Proin condimentum pretium enim. Fusce fringilla, libero et venenatis facilisis, eros enim cursus arcu, vitae facilisis odio augue vitae orci. Aliquam varius nibh ut odio. Sed condimentum condimentum nunc. Pellentesque eget massa. Pellentesque quis mauris. Donec ut ligula ac pede pulvinar lobortis. Pellentesque euismod. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent elit. Ut laoreet ornare est. Phasellus gravida vulputate nulla. Donec sit amet arcu ut sem tempor malesuada. Praesent hendrerit augue in urna. Proin enim ante, ornare vel, consequat ut, blandit in, justo. Donec felis elit, dignissim sed, sagittis ut, ullamcorper a, nulla. Aenean pharetra vulputate odio.

Quisque enim. Proin velit neque, tristique eu, eleifend eget, vestibulum nec, lacus. Vivamus odio. Duis odio urna, vehicula in, elementum aliquam, aliquet laoreet, tellus. Sed velit. Sed vel mi ac elit aliquet interdum. Etiam sapien neque, convallis et, aliquet vel, auctor non, arcu. Aliquam suscipit aliquam lectus. Proin tincidunt magna sed wisi. Integer blandit lacus ut lorem. Sed luctus justo sed enim.

Morbi malesuada hendrerit dui. Nunc mauris leo, dapibus sit amet, vestibulum et, commodo id, est. Pellentesque purus. Pellentesque tristique, nunc ac pulvinar adipiscing, justo eros consequat lectus, sit amet posuere lectus neque vel augue. Cras consectetuer libero ac eros. Ut eget massa. Fusce sit amet enim eleifend sem dictum auctor. In eget risus luctus wisi convallis pulvinar. Vivamus sapien risus, tempor in, viverra in, aliquet pellentesque, eros. Aliquam euismod libero a sem.

Nunc velit augue, scelerisque dignissim, lobortis et, aliquam in, risus. In eu eros. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Curabitur vulputate elit viverra augue. Mauris fringilla, tortor sit amet malesuada mollis, sapien mi dapibus odio, ac imperdiet ligula enim eget nisl. Quisque vitae pede a pede aliquet suscipit. Phasellus tellus pede, viverra vestibulum, gravida id, laoreet in, justo. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer commodo luctus lectus. Mauris justo. Duis varius eros. Sed quam. Cras lacus eros, rutrum eget, varius quis, convallis iaculis, velit. Mauris imperdiet, metus at tristique venenatis, purus neque pellentesque mauris, a ultrices elit lacus nec tortor. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent malesuada. Nam lacus lectus, auctor sit amet, malesuada vel, elementum eget, metus. Duis neque pede, facilisis eget, egestas elementum, nonummy id, neque.

5.2 UML Diagrams

(Space reserved for Major 5 UML diagrams) Proin non sem. Donec nec erat. Proin

Figure 5.1: UML Diagram 1

Figure 5.2: UML Diagram 2

Figure 5.3: UML Diagram 3

Figure 5.4: UML Diagram 4

Figure 5.5: UML Diagram 5

libero. Aliquam viverra arcu. Donec vitae purus. Donec felis mi, semper id, scelerisque porta, sollicitudin sed, turpis. Nulla in urna. Integer varius wisi non elit. Etiam nec sem. Mauris consequat, risus nec congue condimentum, ligula ligula suscipit urna, vitae porta odio erat quis sapien. Proin luctus leo id erat. Etiam massa metus, accumsan pellentesque, sagittis sit amet, venenatis nec, mauris. Praesent urna eros, ornare nec, vulputate eget, cursus sed, justo. Phasellus nec lorem. Nullam ligula ligula, mollis sit amet, faucibus vel, eleifend ac, dui. Aliquam erat volutpat.

Fusce vehicula, tortor et gravida porttitor, metus nibh congue lorem, ut tempus purus mauris a pede. Integer tincidunt orci sit amet turpis. Aenean a metus. Aliquam vestibulum lobortis felis. Donec gravida. Sed sed urna. Mauris et orci. Integer ultrices feugiat ligula. Sed dignissim nibh a massa. Donec orci dui, tempor sed, tincidunt nonummy, viverra sit amet, turpis. Quisque lobortis. Proin venenatis tortor nec wisi. Vestibulum placerat. In hac habitasse platea dictumst. Aliquam porta mi quis risus. Donec sagittis luctus diam. Nam ipsum elit, imperdiet vitae, faucibus nec, fringilla eget, leo. Etiam quis dolor in sapien porttitor imperdiet.

Cras pretium. Nulla malesuada ipsum ut libero. Suspendisse gravida hendrerit tellus. Maecenas quis lacus. Morbi fringilla. Vestibulum odio turpis, tempor vitae, scelerisque a, dictum non, massa. Praesent erat felis, porta sit amet, condimentum sit amet, placerat et, turpis. Praesent placerat lacus a enim. Vestibulum non eros. Ut congue. Donec tristique varius tortor. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam dictum dictum urna.

Phasellus vestibulum orci vel mauris. Fusce quam leo, adipiscing ac, pulvinar eget, molestie sit amet, erat. Sed diam. Suspendisse eros leo, tempus eget, dapibus sit amet, tempus eu, arcu. Vestibulum wisi metus, dapibus vel, luctus sit amet, condimentum quis, leo. Suspendisse molestie. Duis in ante. Ut sodales sem sit amet mauris. Suspendisse ornare pretium orci. Fusce tristique enim eget mi. Vestibulum eros elit, gravida ac, pharetra sed, lobortis in, massa. Proin at dolor. Duis accumsan accumsan pede. Nullam blandit elit in magna lacinia hendrerit. Ut nonummy luctus eros. Fusce eget tortor.

Ut sit amet magna. Cras a ligula eu urna dignissim viverra. Nullam tempor leo porta ipsum. Praesent purus. Nullam consequat. Mauris dictum sagittis dui. Vestibulum sollicitudin consectetuer wisi. In sit amet diam. Nullam malesuada pharetra risus. Proin lacus arcu, eleifend sed, vehicula at, congue sit amet, sem. Sed sagittis pede a nisl. Sed tincidunt odio a pede. Sed dui. Nam eu enim. Aliquam sagittis lacus eget libero. Pellentesque diam sem, sagittis molestie, tristique et, fermentum ornare, nibh. Nulla et tellus non felis imperdiet mattis. Aliquam erat volutpat.

5.3 System Workflow

The workflow begins with historical workload data and proceeds through preprocessing, prediction, resource quantification, allocation, and simulation. The simulation results provide feedback to the reinforcement-learning controller. Vestibulum sodales ipsum id augue. Integer ipsum pede, convallis sit amet, tristique vitae, tempor ut, nunc. Nam non ligula non lorem convallis hendrerit. Maecenas hendrerit. Sed magna odio, aliquam imperdiet, porta ac, aliquet eget, mi. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum nisl sem, dignissim vel, euismod quis,

egestas ut, orci. Nunc vitae risus vel metus euismod laoreet. Cras sit amet neque a turpis lobortis auctor. Sed aliquam sem ac elit. Cras velit lectus, facilisis id, dictum sed, porta rutrum, nisl. Nam hendrerit ipsum sed augue. Nullam scelerisque hendrerit wisi. Vivamus egestas arcu sed purus. Ut ornare lectus sed eros. Suspendisse potenti. Mauris sollicitudin pede vel velit. In hac habitasse platea dictumst.

Suspendisse erat mauris, nonummy eget, pretium eget, consequat vel, justo. Pellentesque consectetuer erat sed lacus. Nullam egestas nulla ac dui. Donec cursus rhoncus ipsum. Nunc et sem eu magna egestas malesuada. Vivamus dictum massa at dolor. Morbi est nulla, faucibus ac, posuere in, interdum ut, sapien. Proin consectetuer pretium urna. Donec sit amet nibh nec purus dignissim mattis. Phasellus vehicula elit at lacus. Nulla facilisi. Cras ut arcu. Sed consectetuer. Integer tristique elit quis felis consectetuer eleifend. Cras et lectus.

Ut congue malesuada justo. Curabitur congue, felis at hendrerit faucibus, mauris lacus porttitor pede, nec aliquam turpis diam feugiat arcu. Nullam rhoncus ipsum at risus. Vestibulum a dolor sed dolor fermentum vulputate. Sed nec ipsum dapibus urna bibendum lobortis. Vestibulum elit. Nam ligula arcu, volutpat eget, lacinia eu, lobortis ac, urna. Nam mollis ultrices nulla. Cras vulputate. Suspendisse at risus at metus pulvinar malesuada. Nullam lacus. Aliquam tempus magna. Aliquam ut purus. Proin tellus.

Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Donec scelerisque metus. Maecenas non mi ut metus porta hendrerit. Nunc semper. Cras quis wisi ut lorem posuere tristique. Nunc vestibulum scelerisque nulla. Suspendisse pharetra sollicitudin ante. Praesent at augue sit amet ante interdum porta. Nunc bibendum augue luctus diam. Etiam nec sem. Sed eros turpis, facilisis nec, vehicula vitae, aliquam sed, nulla. Curabitur justo leo, vestibulum eget, tristique ut, tempus at, nisl.

Nulla venenatis lorem id arcu. Morbi cursus urna a ipsum. Donec porttitor. Integer eleifend, est non mattis malesuada, mi nulla convallis mi, et auctor lectus sapien ut purus. Aliquam nulla augue, pharetra sit amet, faucibus semper, molestie vel, nibh. Pellentesque vestibulum magna et mi. Sed fringilla dolor vel tellus. Nunc libero nunc, venenatis eget, convallis hendrerit, iaculis elementum, mi. Nullam aliquam, felis et accumsan vehicula, magna justo vehicula diam, eu condimentum nisl felis et nunc. Quisque volutpat mauris a velit. Pellentesque massa. Integer at lorem. Nam metus erat, lacinia id, convallis ut, pulvinar non, wisi. Cras iaculis mauris ut neque. Cras sodales, sem vitae imperdiet consequat, pede purus sollicitudin urna, ac aliquam metus orci in leo. Ut molestie ultrices mauris. Vivamus vitae sem. Aliquam erat volutpat. Praesent commodo, nisl ac dapibus aliquet, tortor orci sodales lorem, non ornare nulla lorem quis nisl.

Chapter 6

Project Planning

- The project is planned over 16 weeks. Weeks 1-3 cover problem finalisation, the literature survey, and methodology. Weeks 4-6 are for requirements and the SRS, followed by dataset preparation in Week 7.
- Weeks 8-10 cover CPU and memory prediction and the minimum-resource quantification method. Weeks 11-13 are for setting up CloudSim Plus, developing the RL controller, and integrating the system.
- Weeks 14-15 are reserved for experiments, baseline comparisons, and result analysis. The final week is for completing the report, research paper, and presentation.
- The project will use the Google Cluster Usage Trace, Python and its machine-learning libraries, Java, and CloudSim Plus. A standard development computer is sufficient; a dedicated GPU or paid cloud infrastructure is not required.
- The system will be evaluated against static allocation, reactive threshold-based allocation, and prediction-only allocation. The evaluation will measure prediction error, SLA violations, resource utilisation, wastage, allocation cost, and execution or response time.

Sed at sem vitae purus ultrices vestibulum. Vestibulum tincidunt lacus et ligula. Pellentesque vitae elit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Duis ornare, erat eget laoreet vulputate, lacus ipsum suscipit turpis, et bibendum nisl orci non lectus. Vestibulum nec risus nec libero fermentum fringilla. Morbi non velit in magna gravida hendrerit. Pellentesque quis lectus. Vestibulum eleifend lobortis leo. Vestibulum non augue. Vivamus dictum tempor dui. Maecenas at ligula id felis congue porttitor. Nulla leo magna, egestas quis, vulputate sit amet, viverra id, velit.

Ut lectus lectus, ultricies sit amet, semper eget, laoreet non, ante. Proin at massa quis nunc rhoncus mattis. Aliquam lorem. Curabitur pharetra dui at neque. Aliquam eu tellus. Aenean tempus, felis vitae vulputate iaculis, est dolor faucibus urna, in viverra wisi neque non risus. Fusce vel dolor nec sapien pretium nonummy. Integer faucibus massa ac nulla ornare venenatis. Nulla quis sapien. Sed tortor. Phasellus eget mi. Cras nunc. Cras a enim.

Quisque nisl. In dignissim dapibus massa. Aenean sem magna, scelerisque nec, ullamcorper quis, porttitor ut, lectus. Fusce dignissim facilisis tortor. Vivamus gravida felis sit amet nunc. Nam pulvinar odio vel enim. Pellentesque sit amet est. Vivamus pulvinar leo non sapien. Aliquam erat volutpat. Ut elementum auctor metus. Mauris vestibulum

neque vitae eros. Pellentesque aliquam quam. Donec venenatis tristique purus. In nisl. Nulla velit libero, fermentum at, porta a, feugiat vitae, urna. Etiam aliquet ornare ipsum. Proin non dolor. Aenean nunc ligula, venenatis suscipit, porttitor sit amet, mattis suscipit, magna. Vivamus egestas viverra est. Morbi at risus sed sapien sodales pretium.

Morbi congue congue metus. Aenean sed purus. Nam pede magna, tristique nec, porta id, sollicitudin quis, sapien. Vestibulum blandit. Suspendisse ut augue ac nibh ullamcorper posuere. Integer euismod, neque at eleifend fringilla, augue elit ornare dolor, vel tincidunt purus est id lacus. Vivamus lorem dui, commodo quis, scelerisque eu, tincidunt non, magna. Cras sodales. Quisque vestibulum pulvinar diam. Phasellus tincidunt, leo vitae tristique facilisis, ipsum wisi interdum sem, dapibus semper nulla velit vel lectus. Cras dapibus mauris et augue. Quisque cursus nulla in libero. Suspendisse et lorem sit amet mauris malesuada mollis. Nullam id justo. Maecenas venenatis. Donec lacus arcu, egestas ac, fermentum consectetuer, tempus eu, metus. Proin sodales, sem in pretium fermentum, arcu sapien commodo mauris, venenatis consequat augue urna in wisi. Quisque sapien nunc, varius eget, condimentum quis, lacinia in, est. Fusce facilisis. Praesent nec ipsum.

Suspendisse a dolor. Nam erat eros, congue eget, sagittis a, lacinia in, pede. Maecenas in elit. Proin molestie varius nibh. Vivamus tristique purus sed augue. Proin egestas semper tortor. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Vestibulum orci enim, sagittis ornare, eleifend ut, mattis at, ligula. Nulla molestie convallis arcu. Ut eros tellus, condimentum at, sodales in, ultrices vel, nulla.

Chapter 7

Results and Implementation

7.1 Implementation Screenshots

(Space reserved for Implementation screenshots 1.jpeg to 7.jpeg) Duis magna ante,

Figure 7.1: Code Workspace Creation (Reference 1.jpeg)

Figure 7.2: Resource Optimizer Dashboard (Reference 2.jpeg)

Figure 7.3: IDE Environment (Reference 3.jpeg)

Figure 7.4: Machine Learning Optimizer View (Reference 4.jpeg)

Figure 7.5: Infrastructure Launch View (Reference 5.jpeg)

Figure 7.6: Room Management (Reference 6.jpeg)

Figure 7.7: Login Interface (Reference 7.jpeg)

bibendum eget, eleifend eget, suscipit sed, neque. Vestibulum in mi sed massa cursus cursus. Pellentesque pulvinar mollis neque. Fusce ut enim vitae mauris malesuada tincidunt. Vivamus a neque. Mauris pulvinar, sapien id condimentum dictum, quam arcu rhoncus dui, id tempor lacus justo et justo. Proin sit amet orci eu diam eleifend blandit. Nunc erat massa, luctus ac, fermentum lacinia, tincidunt ultrices, sapien. Praesent sed orci vitae dolor sollicitudin adipiscing. Cras a neque. Ut risus dui, interdum at, placerat id, tristique eu, enim. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Etiam adipiscing eros vestibulum dolor. Pellentesque aliquam, diam eget eleifend posuere, augue eros porttitor lectus, ac dignissim dui metus nec felis. Quisque lacinia. Vestibulum tellus. Suspendisse nec wisi. Aenean ac felis. Aliquam ultrices metus et nulla.

Praesent sed est non nibh tempus venenatis. Praesent rhoncus. Curabitur sagittis est sit amet neque. Sed commodo malesuada lectus. Phasellus enim tellus, tempor ut, tristique eu, aliquam eu, quam. Aenean quis quam quis wisi gravida vehicula. Pellentesque a massa a leo pretium rhoncus. Suspendisse ultrices. Donec lacinia malesuada massa. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec pretium ornare mauris. Phasellus auctor erat eget enim. Integer scelerisque, felis eu consequat fringilla, lorem wisi ultricies velit, id vehicula purus nulla eget odio. Nullam mattis, diam a rutrum fermentum, odio sapien tristique quam, id mollis tellus quam in odio. Mauris eu sapien. Donec aliquam lorem sit amet lorem pharetra lobortis.

Donec ac velit. Sed convallis vestibulum sapien. Vivamus tempor lacus sed lacus. Nunc ut lorem. Ut et tortor. Nullam varius wisi at diam. Etiam ultricies, dolor sit amet fermentum vulputate, neque libero vestibulum orci, vitae fringilla neque arcu aliquet ante. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Quisque venenatis lobortis augue. Sed tempor, tellus iaculis pellentesque pharetra, pede dui malesuada mauris, vel ultrices urna mauris ac nibh. Etiam nibh odio, ultricies vehicula, vestibulum vitae, feugiat eleifend, felis. Vivamus pulvinar. Aliquam erat volutpat. Nulla egestas venenatis metus. Nam feugiat nunc quis elit egestas sagittis. Sed vitae felis. In libero arcu, rhoncus in, commodo eget, auctor in, enim. Vivamus suscipit est. Nulla dapibus, magna vel aliquet egestas, massa massa hendrerit lacus, ac rutrum tellus tellus sit amet felis. Cras viverra.

Suspendisse eu nunc. Aliquam dignissim urna sit amet mauris. Cras commodo, urna ut porttitor venenatis, arcu metus sodales risus, vitae gravida sapien ligula in est. Donec vulputate sollicitudin wisi. Donec vehicula, est id interdum ornare, nibh tellus consectetuer justo, a ultrices felis erat at lectus. In est massa, malesuada non, suscipit at, ullamcorper eu, elit. Nam nulla lacus, bibendum sit amet, sagittis sed, tempor eget, libero. Praesent ligula. Suspendisse nulla. Etiam diam. Nulla ante diam, vestibulum et, aliquet ac, imperdiet vitae, urna. Fusce tincidunt lacus vel elit. Maecenas dictum, tortor non euismod bibendum, pede nibh pretium tellus, at dignissim leo eros eget pede. Nulla venenatis eleifend eros. Aenean ut odio dignissim augue rutrum faucibus. Fusce posuere, tellus eget viverra mattis, erat tellus porta mi, at facilisis sem nibh non urna. Phasellus quis turpis quis mauris suscipit vulputate. Sed interdum lacus non velit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae.

Vivamus vehicula leo a justo. Quisque nec augue. Morbi mauris wisi, aliquet vitae, dignissim eget, sollicitudin molestie, ligula. In dictum enim sit amet risus. Curabitur vitae velit eu diam rhoncus hendrerit. Vivamus ut elit. Praesent mattis ipsum quis turpis. Curabitur rhoncus neque eu dui. Etiam vitae magna. Nam ullamcorper. Praesent interdum bibendum magna. Quisque auctor aliquam dolor. Morbi eu lorem et est porttitor fermentum. Nunc egestas arcu at tortor varius viverra. Fusce eu nulla ut nulla interdum consectetuer. Vestibulum gravida. Morbi mattis libero sed est.

Nam quis enim. Quisque ornare dui a tortor. Fusce consequat lacus pellentesque metus. Duis euismod. Duis non quam. Maecenas vitae dolor in ipsum auctor vehicula. Vivamus nec nibh eget wisi varius pulvinar. Cras a lacus. Etiam et massa. Donec in nisl sit amet dui imperdiet vestibulum. Duis porttitor nibh id eros.

Mauris consectetuer, wisi eu lobortis scelerisque, urna nibh feugiat quam, id congue eros justo eget orci. Ut tellus. Maecenas mattis sapien sed eros. Aliquam quis lectus. Donec nec massa ac turpis semper cursus. Etiam consectetuer ante vel odio. Aliquam tincidunt felis non dolor. Cras id augue ut nisl pretium placerat. Phasellus sapien sapien, pharetra sed, aliquam nec, suscipit a, nibh. Suspendisse risus. Nulla ut mi eget tellus sollicitudin euismod. Vestibulum malesuada malesuada dui. Ut at est ac dui aliquam sagittis. Aliquam erat volutpat.

Curabitur ullamcorper est in mauris. Praesent ac massa. Quisque enim odio, lobortis nec, mattis ut, luctus et, mauris. Mauris eu risus. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Duis eu ligula. Nulla vehicula leo tincidunt erat. Maecenas et nunc. Sed ut sapien. Vestibulum in est. Vestibulum rhoncus.

Donec metus metus, condimentum eu, accumsan nec, vulputate non, purus. Vestibulum ullamcorper vehicula sapien. Mauris risus odio, hendrerit ac, congue ac, ullamcorper at, odio. Aenean leo justo, commodo vitae, placerat blandit, malesuada vel, sem. Donec sit amet ante eget mauris adipiscing sollicitudin. Curabitur posuere sem et leo. Nulla ultricies mauris. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Fusce sollicitudin augue vel tellus. Vivamus mauris eros, pharetra vel, lacinia pretium, egestas a, nibh. Morbi a ligula.

Donec vitae turpis. Suspendisse porttitor. Mauris aliquam purus vitae tellus. Morbi metus diam, tempus ac, cursus ut, ultricies quis, nulla. Praesent nec justo. In lobortis. Donec nec lectus a neque laoreet rhoncus. Quisque in risus nec wisi lacinia ullamcorper. In placerat. Proin facilisis sollicitudin libero. Integer eget neque et pede placerat aliquet. Aliquam purus nulla, pulvinar ut, facilisis quis, sodales sed, magna. Curabitur nulla lectus, rutrum id, bibendum ut, sagittis eget, diam. Sed porta dolor eget est. Integer hendrerit orci. In hac habitasse platea dictumst.

Chapter 8

Conclusion

Cloud resource allocation must respond to changing workloads while balancing application performance with the cost of unused resources. Static and reactive approaches can struggle with unpredictable demand, while workload predictions alone do not directly identify the minimum allocation needed to satisfy an SLA.

This project proposes a simulation-based system that predicts CPU and memory demand, quantifies a minimum feasible resource configuration under a defined SLA condition, and uses the result to guide proactive allocation through reinforcement learning. The system will be implemented and evaluated in CloudSim Plus using historical workload data.

The evaluation will compare the proposed approach with static, reactive threshold-based, and prediction-only baselines. It will measure prediction error, SLA violations, resource utilisation, wastage, allocation cost, and execution or response time. An ablation experiment will also examine the effect of the minimum-resource quantification stage.

The project does not assume in advance that the proposed method will outperform existing approaches. Its effectiveness will be established through the planned experiments. Future work may extend the system to additional resource types or validate it with other workload datasets, subject to time and data availability. Ut facilisis. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Sed pellentesque, turpis sit amet aliquet porta, risus odio venenatis felis, at porta tellus lacus vitae nisl. Donec augue. Quisque consequat, pede laoreet pellentesque posuere, urna sapien tempor justo, eu aliquam tortor nunc id mauris. Fusce pretium, purus facilisis consequat mattis, ligula leo pretium mauris, ac suscipit augue sapien sit amet ipsum. Praesent et ligula eget tortor dapibus blandit. Duis rutrum felis eget dolor. Vestibulum quis elit. Integer dignissim, velit at scelerisque congue, ipsum nulla dignissim dolor, lacinia scelerisque neque erat a mi. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Quisque ipsum lectus, euismod et, lacinia eu, iaculis eu, pede. Etiam justo quam, cursus ut, vulputate vel, feugiat ut, eros. Fusce eleifend mollis ipsum.

Nulla facilisi. Nunc nec elit. Integer ornare convallis tortor. Proin ac diam. In est sapien, laoreet euismod, mattis a, tincidunt at, risus. Vivamus risus. Vestibulum aliquam, urna aliquam porttitor accumsan, nulla tortor ullamcorper elit, ut consequat augue purus sit amet libero. Vivamus nisl lacus, commodo vel, dignissim ut, vestibulum id, pede. Curabitur malesuada hendrerit libero. Mauris quis dolor in tellus varius posuere. Sed vulputate elit at wisi. Fusce vitae neque. Nulla consectetuer, nunc ac eleifend laoreet, mi nulla commodo wisi, vel faucibus ligula lectus ut arcu. Vivamus hendrerit.

Sed varius, nulla vitae tincidunt lobortis, nibh ipsum sollicitudin libero, et commodo

tellus massa in neque. Nulla facilisi. Aenean nec lectus. Aliquam fermentum. Duis ut magna et augue interdum gravida. Morbi elit. Fusce malesuada tempus ipsum. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Mauris iaculis enim non metus. Nullam dui magna, congue et, suscipit sed, aliquam vel, turpis. Quisque ultricies.

Suspendisse feugiat sapien laoreet ante. Integer fringilla, erat eget adipiscing ultrices, nibh dui sollicitudin nunc, in lobortis arcu odio vitae erat. Fusce bibendum ultricies lacus. Mauris eleifend ligula a ante. Etiam faucibus cursus pede. Mauris enim eros, malesuada eu, mattis sit amet, blandit in, nulla. Fusce sit amet purus id mi posuere tincidunt. Mauris sit amet quam vitae quam semper accumsan. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Nam a justo at quam accumsan euismod. Duis tincidunt tristique risus. Ut vel nibh vel libero varius malesuada. In hac habitasse platea dictumst. Morbi sagittis mattis lorem. Pellentesque metus tellus, rutrum vitae, malesuada et, pharetra accumsan, ante. Quisque ac metus ac nisl gravida pellentesque. Sed dapibus feugiat sapien. Vestibulum nec nunc eget sem aliquam lobortis. Suspendisse aliquam quam quis metus.

Suspendisse in odio. In elit diam, cursus vitae, venenatis in, molestie in, leo. Cras ornare. Nulla libero. Phasellus feugiat mattis libero. Sed vehicula aliquam ligula. Nullam lacinia, felis vel dignissim sodales, enim lectus lobortis diam, quis nonummy mauris odio auctor tortor. Integer in dui nec lacus bibendum ultrices. Etiam odio elit, aliquam et, porttitor id, interdum cursus, elit. Nulla eleifend tempor mauris. In vel arcu quis pede laoreet vulputate.

Morbi pharetra magna a lorem. Cras sapien. Duis porttitor vehicula urna. Phasellus iaculis, mi vitae varius consequat, purus nibh sollicitudin mauris, quis aliquam felis dolor vel elit. Quisque neque mi, bibendum non, tristique convallis, congue eu, quam. Etiam vel felis. Quisque ac ligula at orci pulvinar rutrum. Donec mi eros, sagittis eu, consectetuer sed, sagittis sed, lorem. Nunc sed eros. Nullam pellentesque ante quis lectus. Vivamus lacinia, sapien vel fermentum placerat, purus nisl aliquet odio, et porta wisi dui nec nunc. Fusce porta cursus libero.

Quisque eu mi a augue mollis posuere. Donec tincidunt, lorem at vestibulum pulvinar, felis purus nonummy urna, at accumsan purus dui nec leo. Praesent tortor turpis, vehicula in, aliquet ut, dignissim ac, leo. Curabitur sagittis mi id eros. In magna. Sed vitae elit facilisis elit semper sollicitudin. Curabitur convallis tempor nulla. Nullam non turpis a pede sagittis ultrices. Etiam vulputate pede in ligula. Sed a ante id metus pellentesque suscipit. Sed adipiscing justo vitae sapien. Nunc posuere, pede ullamcorper gravida egestas, justo libero tincidunt arcu, vitae pellentesque arcu leo ut mauris. Pellentesque auctor mauris sit amet elit luctus fringilla. Cras sed wisi. Morbi luctus enim vitae tellus. Vivamus venenatis sodales libero.

In hac habitasse platea dictumst. Suspendisse potenti. Nulla pretium sem sit amet nisl. Nulla facilisi. Sed aliquam, turpis sed hendrerit gravida, nunc metus aliquam urna, eget pharetra nibh urna nec lectus. Duis in nisl a nisl commodo facilisis. Nunc placerat risus sed leo. Duis pellentesque porta libero. Praesent et enim. Aenean ullamcorper, ante sit amet fermentum mollis, ligula metus laoreet magna, accumsan accumsan nibh wisi at wisi. Nam tincidunt tempor neque. Maecenas dolor. Donec interdum nisl. Aliquam quam libero, interdum quis, volutpat sed, semper ut, eros. Pellentesque sodales auctor quam. Nullam suscipit massa nec elit. Nullam vulputate.

Aliquam a nulla. Suspendisse suscipit. Etiam lectus ante, interdum sit amet, euismod venenatis, condimentum eu, urna. Etiam at turpis. Cras quis ligula. Cras varius, sapien

non pellentesque bibendum, mauris wisi sodales sem, ac commodo mauris neque non felis. Sed sollicitudin tincidunt arcu. Nullam vel lectus sit amet magna tincidunt tempor. Phasellus a ante. Donec et diam.

Proin sit amet augue. Praesent lacus. Donec a leo. Ut turpis ante, condimentum sed, sagittis a, blandit sit amet, enim. Integer sed elit. In ultricies blandit libero. Proin molestie erat dignissim nulla convallis ultrices. Aliquam in magna. Etiam sollicitudin, eros a sagittis pellentesque, lacus odio volutpat elit, vel tincidunt felis dui vitae lorem. Etiam leo. Nulla et justo.